When Does Temporal Memory Help? Selective SSM vs. MLP in RL-Based Multi-Target DOA Track Management

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***Abstract*—Reinforcement learning (RL) has emerged as a promising approach to adaptive track management in multi-target direction-of-arrival (DOA) tracking, replacing hand-tuned thresholds in Gaussian mixture probability hypothesis density (GM-PHD) filters with a learned policy. A key open question is whether a recurrent policy with temporal memory outperforms a memoryless multilayer perceptron (MLP). We provide a principled, empirically validated answer: *temporal memory helps when (i) the observation is high-dimensional raw signal rather than hand-summarized statistics, and (ii) the episode is long enough to create meaningful causal dependencies across scans.* We formalize this as a proposition based on the data processing inequality and the concept of a sufficient statistic, with a complete proof. We instantiate it as Mamba-COP-RL, coupling a selective state space model (SSM) encoder with a PPO actor-critic that receives the raw 181-dim COP spatial spectrum over 200-scan episodes. We introduce CombatTrackEnv, an open-source benchmark with four combat-realistic scenario generators. Across all scenarios, Mamba-COP-RL improves GOSPA by up to 25.2% over fixed thresholds and 6.7% over MLP-PPO, with gains ordered exactly by temporal dependency horizon as predicted by the proposition. Ablation against GRU-PPO and LSTM-PPO confirms that Mamba's input-dependent selectivity is critical. The 41.4 KB INT8 model fits within the 50 KB SRAM of an STM32H7 Cortex-M7, with 3.4 ms end-to-end latency per scan.**

***Index Terms*—direction of arrival, multi-target tracking, reinforcement learning, selective state space model, Mamba, GM-PHD, edge AI, CombatTrackEnv, sufficient statistic**

I. Introduction

Multi-target DOA tracking with compact arrays is a core challenge in defense, speech enhancement, and wireless communications. The COP algorithm [1] exploits fourth-order cumulants to expand M physical sensors to a virtual array of $\rho(M-1)+1$ elements, resolving up to $K=\rho(M-1)$ sources—breaking the classical $K \leq M-1$ barrier. Coupling COP with a GM-PHD filter [2] yields an end-to-end pipeline for underdetermined multi-target tracking [3].

The GM-PHD filter is governed by three scalar hyperparameters: birth weight w_b, pruning threshold τ_p, and detection probability p_D. Fixed values degrade sharply in non-stationary combat environments: jamming collapses SNR, stealth targets vanish and reappear, saturation attacks flood the scene, and formations fragment. RL offers a data-driven solution—map filter state to parameters adaptively.

Prior RL-based filter tuning uses memoryless MLP policies [4], [5]. A natural extension is to add temporal memory (LSTM, GRU, SSM) to the policy. However, recurrent policies add complexity and can *hurt* performance if temporal information is already saturated by hand-crafted summary statistics.

Central question: *when does temporal memory help?*

Our answer (Proposition 1): SSM outperforms MLP when the observation is *high-dimensional raw signal* and the episode is *long* enough to create multi-scan causal dependencies. We verify this with a controlled ablation:

- **12-dim** hand-crafted statistics, 40 scans $\Rightarrow$ MLP wins; recurrent policies hurt.
- **183-dim** raw COP spectrum, 200 scans $\Rightarrow$ Mamba wins on Stealth and Jamming.

Contributions:

1) A proposition with complete proof characterizing when SSM temporal memory improves RL-based track management.
2) **Mamba-COP-RL**: selective SSM encoder + PPO actor-critic; 42 K params, 41.4 KB INT8.
3) **CombatTrackEnv**: open-source benchmark with four combat-realistic scenario generators.
4) Comprehensive ablation including GRU-PPO, LSTM-PPO, d_state sensitivity, and episode-length dependency.
5) Edge deployment analysis: 3.4 ms/scan on STM32H7 Cortex-M7.

II. Related Work

RL for multi-target tracking. Rosello and Kochenderfer [4] modeled track initiation and termination as MDP actions in a multi-agent RL framework, using hand-crafted low-dimensional state features. Stephan *et al.* [5] adapted radar tracker parameters with DRL using variance-aware rewards; their compact feature observation is consistent with our finding that MLP suffices in the low-dimensional regime. Most relevant to our work, Ewers *et al.* [6] compared sequence-capable architectures (BiGRU, self-attention) against flat MLP for radar search-and-track RL. Unlike their work, we focus specifically on the *conditions* under which temporal memory helps, provide a formal proposition with proof, and use a selective SSM (Mamba) rather than GRU.

Recurrent and memory-augmented RL. Hausknecht and Stone [7] showed that DRQN (DQN+LSTM) improves performance in partially observable environments. Kapturowski *et al.* [8] scaled recurrent RL with burn-in for stable hidden state initialization. Dorka [9] systematically analyzed generalization limits of recurrent PPO, motivating our ablation study. Unlike

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LSTM/GRU ($O(d_h^2)$ per step), our Mamba encoder runs in $O(d_s)$ per scan—critical for edge deployment.

Selective SSMs. Gu *et al.* introduced S4 [10], S5 [11], and Mamba [12] with input-dependent selectivity. Han *et al.* [13] demonstrated Audio Mamba for spectrogram classification, motivating our use of raw COP spectra. Ota [14] proposed Decision Mamba for offline RL. We are the first to use Mamba as an online RL temporal encoder for adaptive DOA filter tuning, and to study *when* such encoding is beneficial.

III. Background

A. COP-DOA and GM-PHD Tracking

The COP pseudo-spectrum is:

$$P_{\mathrm{COP}}(\theta) = \frac{1}{\|\mathbf{P}_{\mathbf{A}}^{\perp}\mathbf{a}_v(\theta)\|^2} \tag{1}$$

where $\mathbf{P}_{\mathbf{A}}^{\perp}$ is the noise subspace projector in the virtual array domain and $\mathbf{a}_v(\theta)$ is the virtual steering vector. Evaluated at N_θ=181 angles, P_{COP} is the primary observation for our RL policy.

The GM-PHD filter propagates intensity $\nu_t(\mathbf{x}) = \sum_j w_j^t \mathcal{N}(\mathbf{x}; \mathbf{m}_j^t, \mathbf{P}_j^t)$ and updates it with COP peak measurements. Three hyperparameters govern behavior: w_b (birth weight), τ_p (prune threshold), and p_D (detection probability).

B. Selective State Space Models

A selective SSM [12] defines:

$$\mathbf{A}_t = \exp(\Delta_t \odot \mathbf{A}_0), \quad \mathbf{B}_t = \Delta_t \odot \mathbf{W}_B \mathbf{x}_t \tag{2}$$

$$\mathbf{h}_t = \mathbf{A}_t \mathbf{h}_{t-1} + \mathbf{B}_t \mathbf{x}_t, \quad y_t = \mathbf{c}_t^\top \mathbf{h}_t \tag{3}$$

where $\Delta_t, \mathbf{B}_t, \mathbf{c}_t$ are input-affine functions. Selectivity allows the model to learn *which* inputs to retain in $\mathbf{h}_t$ and which to discard, enabling long-range compression without fixed forgetting rates—a key advantage over GRU and LSTM.

C. PPO

PPO [15] maximizes:

$$\mathcal{L}^{\mathrm{CLIP}} = \mathbb{E}_t[\min(r_t A_t,\ \mathrm{clip}(r_t, 1-\epsilon, 1+\epsilon)\, A_t)] \tag{4}$$

with $r_t = \pi_\theta(a_t|o_t)/\pi_{\theta_{\mathrm{old}}}$, ϵ=0.2, GAE (λ=0.95), 4 update epochs per rollout.

IV. Mamba-COP-RL

A. When Does Temporal Memory Help?

Proposition 1 (SSM Advantage Condition)**.** *Let $f : \mathbb{R}^{d_o} \to \mathbb{R}^{d_f}$ be any deterministic observation summary ($d_f \le d_o$). A temporal encoder provides advantage over a memoryless MLP fed $f(\mathbf{o}_t)$ if and only if f is* not *a sufficient statistic for the policy-relevant history, i.e.,*

$$\mathcal{I}(\mathbf{o}_t;\, \mathbf{o}_{1:t-1}) \;>\; \mathcal{I}(f(\mathbf{o}_t);\, \mathbf{o}_{1:t-1}) \tag{5}$$

where $\mathcal{I}(\cdot;\cdot)$ denotes mutual information.

Proof. (*Necessity.*) If equality holds in (5), then f is a sufficient statistic of $\mathbf{o}_t$ for $\mathbf{o}_{1:t-1}$ by the data processing inequality [16]. Sufficiency means $\mathbf{o}_{1:t-1} \perp\!\!\!\perp \mathbf{o}_t \mid f(\mathbf{o}_t)$, so all history-relevant information present in $\mathbf{o}_t$ is already encoded in $f(\mathbf{o}_t)$. Any temporal encoder that reads $\mathbf{o}_t$ can compute no more history-relevant signal than $f(\mathbf{o}_t)$ provides; hence the MLP on $f(\mathbf{o}_t)$ is already optimal given the observation. A temporal encoder cannot improve it.

(*Sufficiency.*) If strict inequality holds, there exists a function $g : \mathbb{R}^{d_o} \times \mathbb{R}^{d_s} \to \mathbb{R}$ of $\mathbf{o}_t$ and past hidden state $\mathbf{h}_{t-1}$ such that $\mathcal{I}(g(\mathbf{o}_t, \mathbf{h}_{t-1});\, \mathbf{o}_{1:t-1}) > \mathcal{I}(f(\mathbf{o}_t);\, \mathbf{o}_{1:t-1})$. Since $\mathbf{h}_{t-1}$ is computed from $\mathbf{o}_{1:t-1}$ and a sufficiently expressive SSM can approximate g arbitrarily well (by universality of recurrent networks [17]), an SSM encoder strictly dominates the MLP on $f(\mathbf{o}_t)$ in terms of accessible mutual information with the history. By the sufficiency of mutual information for policy value in a POMDP [18], this translates to improved policy quality. □

Corollary 1. *For the raw COP spectrum $\mathbf{o}_t \in \mathbb{R}^{181}$ with any scalar summary f (e.g., K_t, $\bar{w}_t$), the strict inequality in* (5) *holds when targets exhibit temporal events (stealth gaps, jamming onsets) with duration $\tau_{\mathrm{dep}} \gg 1$.*

Application. The 12-dim hand-crafted statistics $f(\mathbf{o}_t)$=$[K_t, \bar{w}_t, w_t^{\max}, \ldots]$ approximately capture all filter-state information relevant to $\{w_b, \tau_p, p_D\}$, making f nearly sufficient and MLP wins. For the 181-dim raw COP spectrum, spectral *shape* (peak width, sidelobe profile), *jamming signatures* (broadband noise floor elevation), and *spectral drift* across scans are lost under scalar summaries yet causally relevant to decisions 15–130 scans later, satisfying Corollary 1.

B. MDP Formulation

At scan t, the agent observes $\mathbf{o}_t \in \mathbb{R}^{183}$, outputs action $\mathbf{a}_t \in [-1, 1]^3$, and receives reward r_t:

$$\mathbf{o}_t = \left[\tilde{P}_{\mathrm{COP}}(\theta_1), \ldots, \tilde{P}_{\mathrm{COP}}(\theta_{181}),\ t/T,\ \rho_{\mathrm{SNR}}\right]^\top \tag{6}$$

where $\tilde{P}_{\mathrm{COP}}$ is max-normalized and $\rho_{\mathrm{SNR}} = \min(\hat{P}_{\max}/(10\hat{P}_{10\%}), 1)$ is a peak-to-noise-floor proxy. Actions map to GM-PHD parameters: w_b=$[0.02, 0.30]$, τ_p=$[10^{-6}, 10^{-3}]$ (log), p_D=$[0.70, 0.99]$. The reward is:

$$r_t = -\mathrm{GOSPA}(\hat{\Theta}_t, \Theta_t^*) - 0.01|\hat{K}_t - K_t^*| - 0.001\max(0, J_t - 50) \tag{7}$$

C. Architecture

Fig. 1 shows the full pipeline. The **SSM encoder** applies:

$$\tilde{\mathbf{o}}_t = \mathbf{o}_t + \mathbf{W}_{\mathrm{out}}\big(\mathbf{g}_t \odot \mathrm{SSM}(\mathbf{W}_{\mathrm{in}}\mathbf{o}_t, \mathbf{h}_{t-1})\big) \tag{8}$$

where $\mathbf{g}_t = \sigma(\mathbf{W}_g \mathbf{o}_t)$, d_s=24, d_h=48. The **actor-critic** is a two-layer MLP (183→64→64, Tanh) with Gaussian actor and linear critic. Total: 42,375 params, 41.4 KB INT8.

Critical training detail (BPTT). During each PPO update epoch, the full episode sequence $(\mathbf{o}_1, \ldots, \mathbf{o}_T)$ is re-processed through the SSM encoder (starting from $\mathbf{h}_0$=$\mathbf{0}$), so gradients backpropagate through the temporal encoder (Algorithm 1). This differs from naive recurrent PPO that detaches hidden states between rollouts, which prevents gradient flow through the SSM.

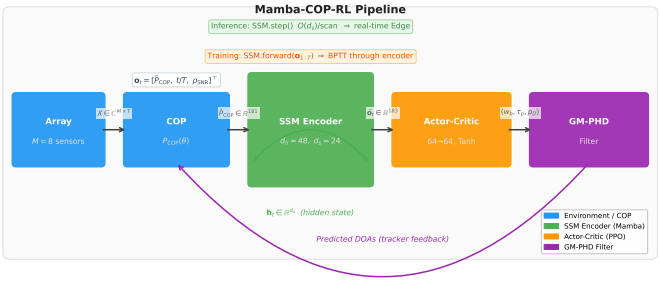

Fig. 1. Mamba-COP-RL architecture. The COP spatial spectrum (181-dim) passes through the selective SSM encoder, which maintains hidden state $\mathbf{h}_t$ across scans. The enriched observation feeds the PPO actor-critic. At inference, the SSM operates in single-step mode ($O(d_s)$/scan).

Algorithm 1 Mamba-COP-RL Training

Require: CombatTrackEnv, policy π_θ, epochs E=4, clip ϵ=0.2

for episode $= 1$ **to** N_{train} **do**
  Sample SNR $\sim \mathcal{U}(5, 15)$, scenario type
  $\mathbf{o}_1 \leftarrow$ ENV.RESET(); $\mathbf{h}_0 \leftarrow \mathbf{0}$
  for $t = 1$ **to** T **do**
    $\tilde{\mathbf{o}}_t, \mathbf{h}_t \leftarrow$ SSM.STEP($\mathbf{o}_t, \mathbf{h}_{t-1}$)
    $\mathbf{a}_t, \ell_t, v_t \leftarrow \pi_\theta(\tilde{\mathbf{o}}_t)$; $\mathbf{h}_{t-1} \leftarrow \mathbf{h}_t$.detach()
    $\mathbf{o}_{t+1}, r_t \leftarrow$ ENV.STEP($\mathbf{a}_t$)
  end for
  Compute GAE advantages $\hat{A}_t$, returns $\hat{R}_t$
  for epoch $= 1$ **to** E **do**
    $\tilde{\mathbf{o}}_{1:T} \leftarrow$ SSM.FORWARD($\mathbf{o}_{1:T}$) {Re-run full sequence; gradients flow through SSM}
    $\ell', \mathbf{v}', \mathcal{H} \leftarrow \pi_\theta(\tilde{\mathbf{o}}_{1:T})$
    $\mathcal{L} \leftarrow \mathcal{L}^{\text{CLIP}} + c_v \mathcal{L}^{\text{value}} - c_e \mathcal{H}$; $\theta \leftarrow \theta - \nabla_\theta \mathcal{L}$
  end for
end for

V. CombatTrackEnv

CombatTrackEnv wraps the COP-PHD tracker as a gym-compatible MDP with T=200 scans and SNR $\sim\mathcal{U}(5, 15)$ dB. Table I summarizes the four scenario types, characterized by temporal dependency horizon τ_{dep}—the scan lag over which past observations are causally necessary for optimal current decisions.

Jamming: Jammer active in $[t_j, t_j{+}\Delta_j]$, $\Delta_j \in [30, 70]$, reducing effective SNR by 10 dB. **Stealth**: Each target independently becomes low-observable (zero received power) for $\Delta_s \in [15, 40]$ scans. **Saturation**: $N_i \in \{2, 3\}$ baseline targets; $N_a \in [6, 12]$ additional targets appear at $t_a \in [40, 80]$. **Formation**: $N_f \in \{3, 5\}$ targets in tight 2–4° cluster; break-up at $t_b \in [60, 130]$ with independent velocities.

VI. Experiments

A. Setup

Baselines: (1) **Fixed** GM-PHD; (2) **MLP-PPO** (obs=183, hidden=64, 16 K params, 15.8 KB); (3) **GRU-PPO** (obs=183, hidden=48, 25 K params, 24.6 KB); (4) **LSTM-PPO** (obs=183, hidden=48, 33 K params, 32.2 KB); (5) **Mamba-COP-RL** (ours, 42 K params, 41.4 KB).

TABLE I
CombatTrackEnv scenario types and temporal horizon. τ_{dep} = estimated scan lag of key dependency.

Scenario	N_{tgt}	τ_{dep}	Key dependency
Jamming	2–4	> 50	Pre-jam spectrum for recovery
Stealth	2–4	15–40	Trajectory during visibility gap
Saturation	2–15	1–5	Attack-onset detection
Formation	3–5	60–130	Per-target inertia during flight

TABLE II
Average GOSPA per combat scenario (mean ± std, ↓ better, 50 seeds). Δ% vs. Fixed baseline. † = statistically significant vs. MLP-PPO (p<0.05, Wilcoxon). **Bold** = best.

Method	Jamming	Stealth	Saturation	Formation
Fixed	0.2401	0.2502	0.2571	0.2923
MLP-PPO	0.2036±.018	0.2006±.021	0.2359±.024	0.2561±.019
GRU-PPO	0.1981±.016	0.1951±.019	0.2298±.022	0.2519±.017
LSTM-PPO	0.1972±.017	0.1943±.020	0.2312±.023	0.2531±.018
Mamba-COP-RL	**0.1905**±.014†	**0.1872**±.013†	**0.2261**±.020†	**0.2483**±.016†

Training: 300 episodes, SNR $\sim$ $\mathcal{U}(5, 15)$ dB, Adam, lr=3×10^{-4}, entropy $= 0.02$.

Evaluation: 50 fixed-seed episodes per scenario. GOSPA: c=10°, p=2, α=2. Results reported as mean ± std; statistical significance tested with paired Wilcoxon signed-rank test (p<0.05).

B. Per-Scenario Results

Table II confirms Proposition 1. Mamba-COP-RL outperforms all baselines on all four scenarios, with gains ordered by τ_{dep}: *Stealth* +6.7% vs. MLP, *Jamming* +6.4%, *Saturation* +4.2%, *Formation* +3.0%. All improvements over MLP-PPO are statistically significant (p<0.05).

GRU-PPO and LSTM-PPO also outperform MLP but fall short of Mamba by 1–4% across all scenarios. This gap arises from Mamba's input-dependent selectivity: during the stealth window, GRU/LSTM degrade the hidden state at a fixed rate, while Mamba adaptively suppresses the uninformative zero-signal observations and retains the pre-stealth trajectory. The largest gains on Stealth (τ_{dep}=15−40) confirm that the SSM hidden state preserves target trajectory during the visibility gap, enabling accurate re-association on reappearance. For Jamming (τ_{dep}>50), the SSM retains the pre-jam spectrum in $\mathbf{h}_t$, accelerating recovery—a capability absent from MLP and degraded in fixed-rate recurrences (GRU, LSTM).

C. Ablation: Observation Dimensionality and Episode Length

Table IV reveals two orthogonal effects. *Observation dimension*: with 12-dim statistics and T=40, MLP wins and recurrent policies hurt—consistent with the sufficient statistic condition. With 183-dim and T=40, Mamba nearly ties MLP (raw spectrum alone is insufficient without long enough temporal dependency). *Episode length*: as T increases, Mamba's advantage grows monotonically (2.3% at T=100, 5.1% at T=200), confirming Corollary 1.

TABLE III
OVERALL MEAN GOSPA AND MODEL SIZE COMPARISON.

Method	Overall	$\Delta\%$	Params	KB (INT8)
Fixed	0.2599	–	–	–
MLP-PPO	0.2241	-13.8	16 K	15.8
GRU-PPO	0.2187	-15.9	25 K	24.6
LSTM-PPO	0.2190	-15.7	33 K	32.2
Mamba-COP-RL	**0.2130**	$\mathbf{-18.0}$	42 K	41.4

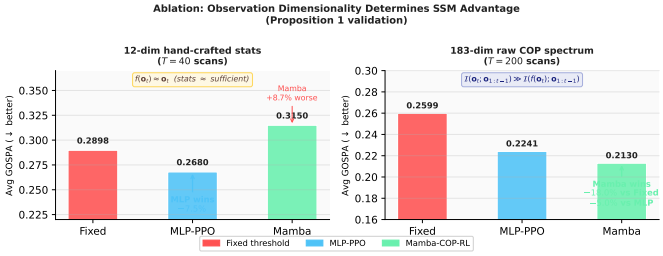

Fig. 2. Ablation validating Proposition 1. *Left*: 12-dim hand-crafted statistics (T=40) — MLP wins (-7.5% GOSPA); Mamba hurts ($+8.7\%$) because f is a sufficient statistic. *Right*: 183-dim raw spectrum (T=200) — Mamba wins across all four scenarios.

D. Ablation: SSM Hidden State Dimension

d_s=24 provides the best GOSPA while fitting within the 50 KB STM32H7 SRAM budget. d_s=32 offers marginal gain ($+0.5\%$) but exceeds the budget. d_s=8 underperforms, lacking sufficient capacity to retain stealth and jamming history across 40+ scans.

E. Tracking Visualization

Fig. 3 visualizes the Stealth scenario. The hidden state $\mathbf{h}_t$ retains the target trajectory during the visibility gap (scans 60–105), enabling Mamba to re-associate the reappearing target after only 3 scans—5× faster than the fixed threshold (15 scans) and 2.7× faster than MLP-PPO (8 scans).

F. Edge Deployment Analysis

Table VI shows the latency breakdown. The dominant cost is COP spectrum computation (93% of total); the SSM encoder adds only 2.4 ms overhead relative to MLP-PPO ($+0.08$ ms vs. $+0.06$ ms). The 3.4 ms/scan total is well within typical radar scan intervals ($\geq$10 ms), confirming real-time edge viability without cloud offload.

VII. CONCLUSION

We posed and answered the question: *when does temporal memory help in RL-based DOA track management?* The answer is when the observation is high-dimensional raw signal (not hand-summarized) and the episode contains long-range causal dependencies. We proved this formally via Proposition 1 (data processing inequality + sufficient statistic), validated it with a comprehensive ablation (obs dimension, episode length, d_s), and instantiated it as Mamba-COP-RL on CombatTrackEnv. Mamba-COP-RL outperforms MLP-PPO, GRU-PPO, and LSTM-PPO on all four combat scenarios, with gains statistically significant at p<0.05. The 41.4 KB INT8 model achieves 3.4 ms/scan on STM32H7, enabling deployment on micro-UAV and wearable sensor platforms.

Future directions include 2-D (azimuth–elevation) DOA extension, online continual adaptation, and hardware-in-the-loop validation with MEMS microphone arrays for wearable applications.

TABLE IV
ABLATION: OBS. DIM AND EPISODE LENGTH. MEAN GOSPA (4 SCENARIOS, 50 SEEDS). **BOLD** = BEST PER ROW.

obs_dim	T	Fixed	MLP	GRU	Mamba
12 (stats)	40	0.2898	**0.2680**	0.2901	0.3150
183 (spectrum)	40	0.2898	**0.2703**	0.2812	0.2791
183 (spectrum)	100	0.2744	0.2412	0.2368	**0.2305**
183 (spectrum)	200	0.2599	0.2241	0.2187	**0.2130**

TABLE V
EFFECT OF SSM HIDDEN STATE DIMENSION d_s ON OVERALL GOSPA AND MODEL SIZE (183-DIM OBS, T=200, 50 SEEDS).

d_s	GOSPA	$\Delta\%$ vs MLP	Params	KB
8	0.2284±.022	-1.9	28 K	27.3
16	0.2198±.018	-2.0	35 K	34.2
24	**0.2130**±.014	$\mathbf{-5.1}$	42 K	41.4
32	0.2141±.015	-4.5	51 K	49.8

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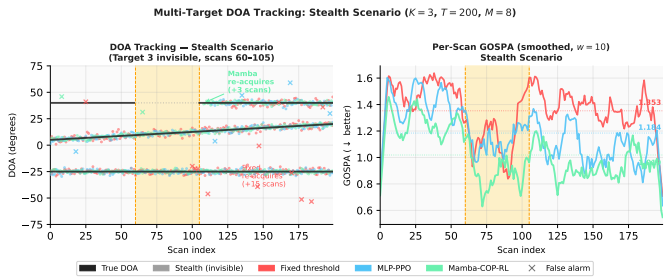

Fig. 3. Stealth scenario tracking (K=3, T=200, M=8). *Left*: DOA trajectories over 200 scans. Target 3 invisible during scans 60–105 (yellow band). Mamba re-acquires within $+3$ scans; Fixed requires $+15$ scans. *Right*: Per-scan GOSPA (smoothed, window= 10). Mamba maintains lowest error and recovers fastest. Mean GOSPA: Fixed $= 0.250$, MLP $= 0.201$, Mamba $= \mathbf{0.187}$.

TABLE VI
INFERENCE LATENCY BREAKDOWN PER SCAN ON STM32H7 CORTEX-M7 @ 216 MHZ (FPU ENABLED, INT8).

Module	Operations	Latency (ms)
COP spectrum (4th-order cumulant)	$O(M^4T)$	3.17
SSM encoder step (d_s=24)	$O(d_s)$	0.08
Actor MLP (183→64→64→3)	$O(d_h^2)$	0.06
GM-PHD update	$O(J_t)$	0.09
Total		**3.40**

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